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Inventor(s)	Donderici; Burkay

Online learning by an instance of a deep learning model and sharing of learning with additional instances of the deep learning model

Abstract

The subject disclosure relates to techniques for enabling sharing of knowledge among a fleet of autonomous vehicles. A process of the disclosed technology can include generating an update for a continuous deep learning neural network on-board the autonomous vehicle based on driving scenarios encountered by the autonomous vehicle during its deployment and providing the update for the continuous deep learning neural network to additional vehicles in the fleet on autonomous vehicles, wherein the update for the continuous deep learning neural network is configured to be incorporated into a joint kernel for use by the additional vehicles in the fleet on autonomous vehicles.

Inventors:	Donderici; Burkay (Burlingame, CA)
Applicant:	GM Cruise Holdings LLC (San Francisco, CA)
Family ID:	1000008782355
Assignee:	GM CRUISE HOLDINGS LLC (San Francisco, CA)
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Primary Examiner: Woolwine; Shane D

Attorney, Agent or Firm: CANTOR COLBURN LLP

Background/Summary

BACKGROUND

1. Technical Field

(1) The subject technology pertains to online learning by an instance of a deep learning model and sharing the of learning with additional instances of the deep learning model deployed on other computing environments, and more particularly to enabling sharing of knowledge learned by an instance of a deep learning model on one vehicle in a fleet with additional instances of the deep learning model deployed on other vehicles of a fleet whereby all instances of the deep learning model can update a joint kernel with the learnings from all instances of the deep learning model.

2. Introduction

(2) Autonomous vehicles are vehicles having computers and control systems that perform driving and navigation tasks that are conventionally performed by a human driver. As autonomous vehicle technologies continue to advance, ridesharing services will increasingly utilize autonomous vehicles to improve service efficiency and safety. However, autonomous vehicles will be required

to perform many of the functions conventionally performed by human drivers, such as avoiding dangerous or difficult routes, and performing other navigation and routing tasks necessary to provide safe and efficient transportation. Such tasks may require the collection and processing of large quantities of data disposed on the autonomous vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The embodiments herein may be better understood by referring to the following description in conjunction with the accompanying drawings in which like reference numerals indicate identically or functionally similar elements, of which:
- (2) FIG. 1 illustrates an example of a system for managing one or more Autonomous Vehicles (AVs) in accordance with some aspects of the present technology.
- (3) FIG. 2 illustrates an example diagram of a Continuous Learning Machine (CLM) for resolving uncommon scenarios in an AV in accordance with some aspects of the present technology.
- (4) FIG. 3 illustrates an example lifecycle of a machine learning (ML) model in accordance with some aspects of the present technology.
- (5) FIG. 4 illustrates an environment having a fleet of autonomous vehicles in accordance with some aspects of the present technology.
- (6) FIG. 5 is a flowchart of a method for sharing knowledge learned by an autonomous vehicle of a fleet in accordance with some aspects of the present technology.
- (7) FIG. 6 is a flowchart of a method for utilizing shared knowledge by an autonomous vehicle of a fleet in accordance with some aspects of the present technology.
- (8) FIG. 7 shows an example of a system for implementing certain aspects of the present technology.

DETAILED DESCRIPTION

- (9) Autonomous vehicles are vehicles having computers and control systems that perform driving and navigation tasks that a human driver conventionally performs. As autonomous vehicle technologies advance, ridesharing services will increasingly utilize autonomous vehicles to improve service efficiency and safety. However, autonomous vehicles will be required to perform many of the functions conventionally performed by human drivers, such as avoiding dangerous or difficult routes and performing other navigation and routing tasks necessary to provide safe and efficient transportation. Such tasks may require the collection and processing of large quantities of data disposed on the autonomous vehicle.
- (10) These autonomous vehicles have a limited amount of time per charge to navigate around routes to perceive objects through their sensors and learn therefrom. Additionally, the storage costs for sensor data can be prohibitively expensive, especially when scaling across a fleet of autonomous vehicles. In other words, there is a limited amount of data that a single autonomous vehicle can record during a trip, and as the amount of stored data increases, the more expensive storage of the data is.
- (11) Thus, the present technology addresses the need for an efficient process for continuous learning by each autonomous vehicle in the fleet of autonomous vehicles. More specifically, the present technology can learn over a network by utilizing an on-board continuous learning model and providing an update to the autonomous vehicle. The learnings can be limited to ensure safety of the autonomous vehicle after the update. Additionally, the learnings from each autonomous vehicle can be gathered and collected into a common stack periodically. For example, each autonomous vehicle can learn and update a kernel of a deep-learning model on-board the respective autonomous vehicle. Then, kernels of a fleet of autonomous vehicles can be averaged out or distilled (e.g., using knowledge distillation methods) into a joint kernel. In other words,

autonomous vehicles can continuously learn and make changes to the deep-learning models on-board the autonomous vehicle while driving. The changes can then be periodically distilled into a single deep-learning model including learnings from a fleet of autonomous vehicles. Thus, the storage of data for each autonomous vehicle can be streamlined into storage for a single deep-learning model that is leveraged by the autonomous vehicles in the fleet.

(12) A description of an AV management system and a continual learning machine (CLM) for the AV management system, as illustrated in FIGS. 1 and 2. An overview of a neural network lifecycle is disclosed in FIG. 3. An environment having a fleet of autonomous vehicles is disclosed in FIG. 4 and is followed by methods performed by the autonomous vehicles in FIGS. 5 and 6. The discussion then concludes with a brief description of example devices, as illustrated in FIG. 7. These variations shall be described herein as the various embodiments are set forth. The disclosure now turns to FIG. 1.

(13) FIG. 1 illustrates an example of an autonomous vehicle (AV) management system **100**. One of ordinary skill in the art will understand that, for the AV management system **100** and any system discussed in the present disclosure, there can be additional or fewer components in similar or alternative configurations. The illustrations and examples provided in the present disclosure are for conciseness and clarity. Other embodiments may include different numbers and/or types of elements, but one of ordinary skill the art will appreciate that such variations do not depart from the scope of the present disclosure.

(14) In this example, the AV management system **100** includes an AV **102**, a data center **150**, and a client computing device **170**. The AV **102**, the data center **150**, and the client computing device **170** can communicate with one another over one or more networks (not shown), such as a public network (e.g., the Internet, an Infrastructure as a Service (IaaS) network, a Platform as a Service (PaaS) network, a Software as a Service (SaaS) network, other Cloud Service Provider (CSP) network, etc.), a private network (e.g., a Local Area Network (LAN), a private cloud, a Virtual Private Network (VPN), etc.), and/or a hybrid network (e.g., a multi-cloud or hybrid cloud network, etc.).

(15) The AV **102** can navigate roadways without a human driver based on sensor signals generated by multiple sensor systems **104**, **106**, and **108**. The sensor systems **104-108** can include different types of sensors and can be arranged about the AV **102**. For instance, the sensor systems **104-108** can comprise Inertial Measurement Units (IMUs), cameras (e.g., still image cameras, video cameras, etc.), light sensors (e.g., light detection and ranging (LIDAR) systems, ambient light sensors, infrared sensors, etc.), RADAR systems, global positioning system (GPS) receivers, audio sensors (e.g., microphones, Sound Navigation and Ranging (SONAR) systems, ultrasonic sensors, etc.), engine sensors, speedometers, tachometers, odometers, altimeters, tilt sensors, impact sensors, airbag sensors, seat occupancy sensors, open/closed door sensors, tire pressure sensors, rain sensors, and so forth. For example, the sensor system **104** can be a camera system, the sensor system **106** can be a LIDAR system, and the sensor system **108** can be a RADAR system. Other embodiments may include any other number and type of sensors.

(16) The AV **102** can also include several mechanical systems that can be used to maneuver or operate the AV **102**. For instance, the mechanical systems can include a vehicle propulsion system **130**, a braking system **132**, a steering system **134**, a safety system **136**, and a cabin system **138**, among other systems. The vehicle propulsion system **130** can include an electric motor, an internal combustion engine, or both. The braking system **132** can include an engine brake, brake pads, actuators, and/or any other suitable componentry configured to assist in decelerating the AV **102**. The steering system **134** can include suitable componentry configured to control the direction of movement of the AV **102** during navigation. The safety system **136** can include lights and signal indicators, a parking brake, airbags, and so forth. The cabin system **138** can include cabin temperature control systems, in-cabin entertainment systems, and so forth. In some embodiments, the AV **102** might not include human driver actuators (e.g., steering wheel, handbrake, foot brake

pedal, foot accelerator pedal, turn signal lever, window wipers, etc.) for controlling the AV **102**. Instead, the cabin system **138** can include one or more client interfaces (e.g., Graphical User Interfaces (GUIs), Voice User Interfaces (VUIs), etc.) for controlling certain aspects of the mechanical systems **130-138**.

(17) The AV **102** can additionally include a local computing device **110** that is in communication with the sensor systems **104-108**, the mechanical systems **130-138**, the data center **150**, and the client computing device **170**, among other systems. The local computing device **110** can include one or more processors and memory, including instructions that can be executed by the one or more processors. The instructions can make up one or more software stacks or components responsible for controlling the AV **102**; communicating with the data center **150**, the client computing device **170**, and other systems; receiving inputs from riders, passengers, and other entities within the AV's environment; logging metrics collected by the sensor systems **104-108**; and so forth. In this example, the local computing device **110** includes a perception stack **112**, a mapping and localization stack **114**, a prediction stack **116**, a planning stack **118**, a communications stack **120**, a control stack **122**, an AV operational database **124**, and a high definition (HD) geospatial database **126**, among other stacks and systems.

(18) The perception stack **112** can enable the AV **102** to “see” (e.g., via cameras, LIDAR sensors, infrared sensors, etc.), “hear” (e.g., via microphones, ultrasonic sensors, RADAR, etc.), and “feel” (e.g., pressure sensors, force sensors, impact sensors, etc.) its environment using information from the sensor systems **104-108**, the mapping and localization stack **114**, the HD geospatial database **126**, other components of the AV, and other data sources (e.g., the data center **150**, the client computing device **170**, third party data sources, etc.). The perception stack **112** can detect and classify objects and determine their current locations, speeds, directions, and the like. In addition, the perception stack **112** can determine the free space around the AV **102** (e.g., to maintain a safe distance from other objects, change lanes, park the AV, etc.). The perception stack **112** can also identify environmental uncertainties, such as where to look for moving objects, flag areas that may be obscured or blocked from view, and so forth. In some embodiments, an output of the prediction stack can be a bounding area around a perceived object that can be associated with a semantic label that identifies the type of object that is within the bounding area, the kinematic of the object (information about its movement), a tracked path of the object, and a description of the pose of the object (its orientation or heading, etc.). The bounding area may be defined on a grid that consists of a rectangular, cylindrical or spherical projection of the camera or LIDAR data.

(19) The mapping and localization stack **114** can determine the AV's position and orientation (pose) using different methods from multiple systems (e.g., GPS, IMUS, cameras, LIDAR, RADAR, ultrasonic sensors, the HD geospatial database **126**, etc.). For example, in some embodiments, the AV **102** can compare sensor data captured in real-time by the sensor systems **104-108** to data in the HD geospatial database **126** to determine its precise (e.g., accurate to the order of a few centimeters or less) position and orientation. The AV **102** can focus its search based on sensor data from one or more first sensor systems (e.g., GPS) by matching sensor data from one or more second sensor systems (e.g., LIDAR). If the mapping and localization information from one system is unavailable, the AV **102** can use mapping and localization information from a redundant system and/or from remote data sources.

(20) The prediction stack **116** can receive information from the localization stack **114** and objects identified by the perception stack **112** and predict a future path for the objects. In some embodiments, the prediction stack **116** can output several likely paths that an object is predicted to take along with a probability associated with each path. For each predicted path, the prediction stack **116** can also output a range of points along the path corresponding to a predicted location of the object along the path at future time intervals along with an expected error value for each of the points that indicates a probabilistic deviation from that point. In some embodiments, the prediction stack **116** can output a probability distribution of likely paths or positions that the object is

predicted to take.

(21) The planning stack **118** can determine how to maneuver or operate the AV **102** safely and efficiently in its environment. For example, the planning stack **118** can receive the location, speed, and direction of the AV **102**, geospatial data, data regarding objects sharing the road with the AV **102** (e.g., pedestrians, bicycles, vehicles, ambulances, buses, cable cars, trains, traffic lights, lanes, road markings, etc.) or certain events occurring during a trip (e.g., emergency vehicle blaring a siren, intersections, occluded areas, street closures for construction or street repairs, double-parked cars, etc.), traffic rules and other safety standards or practices for the road, user input, and other relevant data for directing the AV **102** from one point to another and outputs from the perception stack **112**, localization stack **114**, and prediction stack **116**. The planning stack **118** can determine multiple sets of one or more mechanical operations that the AV **102** can perform (e.g., go straight at a specified rate of acceleration, including maintaining the same speed or decelerating; turn on the left blinker, decelerate if the AV is above a threshold range for turning, and turn left; turn on the right blinker, accelerate if the AV is stopped or below the threshold range for turning, and turn right; decelerate until completely stopped and reverse; etc.), and select the best one to meet changing road conditions and events. If something unexpected happens, the planning stack **118** can select from multiple backup plans to carry out. For example, while preparing to change lanes to turn right at an intersection, another vehicle may aggressively cut into the destination lane, making the lane change unsafe. The planning stack **118** could have already determined an alternative plan for such an event. Upon its occurrence, it could help direct the AV **102** to go around the block instead of blocking a current lane while waiting for an opening to change lanes.

(22) The control stack **122** can manage the operation of the vehicle propulsion system **130**, the braking system **132**, the steering system **134**, the safety system **136**, and the cabin system **138**. The control stack **122** can receive sensor signals from the sensor systems **104-108** as well as communicate with other stacks or components of the local computing device **110** or a remote system (e.g., the data center **150**) to effectuate operation of the AV **102**. For example, the control stack **122** can implement the final path or actions from the multiple paths or actions provided by the planning stack **118**. This can involve turning the routes and decisions from the planning stack **118** into commands for the actuators that control the AV's steering, throttle, brake, and drive unit.

(23) The communications stack **120** can transmit and receive signals between the various stacks and other components of the AV **102** and between the AV **102**, the data center **150**, the client computing device **170**, and other remote systems. The communications stack **120** can enable the local computing device **110** to exchange information remotely over a network, such as through an antenna array or interface that can provide a metropolitan WIFI network connection, a mobile or cellular network connection (e.g., Third Generation (3G), Fourth Generation (4G), Long-Term Evolution (LTE), 5th Generation (5G), etc.), and/or other wireless network connection (e.g., License Assisted Access (LAA), Citizens Broadband Radio Service (CBRS), MULTEFIRE, etc.). The communications stack **120** can also facilitate the local exchange of information, such as through a wired connection (e.g., a user's mobile computing device docked in an in-car docking station or connected via Universal Serial Bus (USB), etc.) or a local wireless connection (e.g., Wireless Local Area Network (WLAN), Bluetooth®, infrared, etc.).

(24) The HD geospatial database **126** can store HD maps and related data of the streets upon which the AV **102** travels. In some embodiments, the HD maps and related data can comprise multiple layers, such as an areas layer, a lanes and boundaries layer, an intersections layer, a traffic controls layer, and so forth. The areas layer can include geospatial information indicating geographic areas that are drivable (e.g., roads, parking areas, shoulders, etc.) or not drivable (e.g., medians, sidewalks, buildings, etc.), drivable areas that constitute links or connections (e.g., drivable areas that form the same road) versus intersections (e.g., drivable areas where two or more roads intersect), and so on. The lanes and boundaries layer can include geospatial information of road lanes (e.g., lane centerline, lane boundaries, type of lane boundaries, etc.) and related attributes

(e.g., direction of travel, speed limit, lane type, etc.). The lanes and boundaries layer can also include 3D attributes related to lanes (e.g., slope, elevation, curvature, etc.). The intersections layer can include geospatial information of intersections (e.g., crosswalks, stop lines, turning lane centerlines and/or boundaries, etc.) and related attributes (e.g., permissive, protected/permissive, or protected only left turn lanes; legal or illegal u-turn lanes; permissive or protected only right turn lanes; etc.). The traffic controls layer can include geospatial information of traffic signal lights, traffic signs, and other road objects and related attributes.

(25) The AV operational database **124** can store raw AV data generated by the sensor systems **104-108**, stacks **112-122**, and other components of the AV **102** and/or data received by the AV **102** from remote systems (e.g., the data center **150**, the client computing device **170**, etc.). In some embodiments, the raw AV data can include HD LIDAR point cloud data, image data, RADAR data, GPS data, and other sensor data that the data center **150** can use for creating or updating AV geospatial data or for creating simulations of situations encountered by AV **102** for future testing or training of various machine learning algorithms that are incorporated in the local computing device **110**.

(26) The data center **150** can be a private cloud (e.g., an enterprise network, a co-location provider network, etc.), a public cloud (e.g., an IaaS network, a PaaS network, a SaaS network, or other CSP network), a hybrid cloud, a multi-cloud, and so forth. The data center **150** can include one or more computing devices remote to the local computing device **110** for managing a fleet of AVs and AV-related services. For example, in addition to managing the AV **102**, the data center **150** may also support a ridesharing service, a delivery service, a remote/roadside assistance service, street services (e.g., street mapping, street patrol, street cleaning, street metering, parking reservation, etc.), and the like.

(27) The data center **150** can send and receive various signals to and from the AV **102** and the client computing device **170**. These signals can include sensor data captured by the sensor systems **104-108**, roadside assistance requests, software updates, ridesharing pick-up and drop-off instructions, and so forth. In this example, the data center **150** includes a data management platform **152**, an Artificial Intelligence/Machine Learning (AI/ML) platform **154**, a simulation platform **156**, a remote assistance platform **158**, and a ridesharing platform **160**, among other systems.

(28) The data management platform **152** can be a “big data” system capable of receiving and transmitting data at high velocities (e.g., near real-time or real-time), processing a large variety of data and storing large volumes of data (e.g., terabytes, petabytes, or more of data). The varieties of data can include data having different structured (e.g., structured, semi-structured, unstructured, etc.), data of different types (e.g., sensor data, mechanical system data, ridesharing service, map data, audio, video, etc.), data associated with different types of data stores (e.g., relational databases, key-value stores, document databases, graph databases, column-family databases, data analytic stores, search engine databases, time series databases, object stores, file systems, etc.), data originating from different sources (e.g., AVs, enterprise systems, social networks, etc.), data having different rates of change (e.g., batch, streaming, etc.), or data having other heterogeneous characteristics. The various platforms and systems of the data center **150** can access data stored by the data management platform **152** to provide their respective services.

(29) The AI/ML platform **154** can provide the infrastructure for training and evaluating machine learning algorithms for operating the AV **102**, the simulation platform **156**, the remote assistance platform **158**, the ridesharing platform **160**, and other platforms and systems. Using the AI/ML platform **154**, data scientists can prepare data sets from the data management platform **152**; select, design, and train machine learning models; evaluate, refine, and deploy the models; maintain, monitor, and retrain the models; and so on.

(30) The simulation platform **156** can enable testing and validation of the algorithms, machine learning models, neural networks, and other development efforts for the AV **102**, the remote assistance platform **158**, the ridesharing platform **160**, and other platforms and systems. The

simulation platform **156** can replicate a variety of driving environments and/or reproduce real-world scenarios from data captured by the AV **102**, including rendering geospatial information and road infrastructure (e.g., streets, lanes, crosswalks, traffic lights, stop signs, etc.) obtained from a cartography platform; modeling the behavior of other vehicles, bicycles, pedestrians, and other dynamic elements; simulating inclement weather conditions, different traffic scenarios; and so on.

(31) The remote assistance platform **158** can generate and transmit instructions regarding the operation of the AV **102**. For example, in response to an output of the AI/ML platform **154** or other system of the data center **150**, the remote assistance platform **158** can prepare instructions for one or more stacks or other components of the AV **102**.

(32) The ridesharing platform **160** can interact with a customer of a ridesharing service via a ridesharing application **172** executing on the client computing device **170**. The client computing device **170** can be any type of computing system, including a server, desktop computer, laptop, tablet, smartphone, smart wearable device (e.g., smartwatch, smart eyeglasses or other Head-Mounted Display (HMD), smart ear pods, or other smart in-ear, on-ear, or over-ear device, etc.), gaming system, or other general purpose computing device for accessing the ridesharing application **172**. The client computing device **170** can be a customer's mobile computing device or a computing device integrated with the AV **102** (e.g., the local computing device **110**). The ridesharing platform **160** can receive requests to pick up or drop off from the ridesharing application **172** and dispatch the AV **102** for the trip.

(33) FIG. 2 illustrates an example diagram of a CLM **200** that solves a long-tail prediction problem in an AV in accordance with some examples. The CLM **200** is a continual loop that iterates and improves based on continual feedback to learn and resolve driving situations experienced by the AV.

(34) The CLM **200** begins with a fleet of AVs outfitted with sensors to record a real-world driving scene. In some cases, the fleet of AVs is situated in a suitable environment that represents challenging and diverse situations such as an urban environment to provide more learning opportunities. The AVs record the driving situations into a collection of driving data **210**.

(35) The CLM **200** includes an error mining **220** service to mine for errors and uses active learning to automatically identify error cases and scenarios having a significant difference between prediction and reality, which are added to a dataset of error instances **230**. The error instances are long-tail scenarios that are uncommon and provide rich examples for simulation and training. The error instances **230** store high-value data and prevent storing datasets with situations that are easily resolved.

(36) The CLM **200** also implements a labeling function **240** that includes both automated and manual data annotation of data stored in error augmented training data **250** and used for future predictions. The automated data annotation is performed by an ML labeling annotator that uses a neural network trained to identify and label error scenarios in the datasets. Using the ML labeling annotator enables significant scale, cost, and speed improvements that allow the CLM **200** to cover more scenarios of the long tail. The labeling function **240** also includes functionality to allow a human annotator to supplement the ML labeling function. By having both an automated ML labeling function and a manual (human) labeling annotator, the CLM **200** can be populated with dense and accurate datasets for prediction.

(37) The final step of the CLM **200** is model training and evaluation **260**. A new model (e.g., a neural network) is trained based on the error augmented training data **250**, and the new model is tested extensively using various techniques to ensure that the new model exceeds the performance of the previous model and generalizes well to the nearly infinite variety of scenarios found in the various datasets. The model can also be simulated in a virtual environment and analyzed for performance. Once the new model has been accurately tested, the new model can be deployed in an AV to record driving data **210**. The CLM **200** is a continual feedback loop that provides continued growth and learning to provide accurate models for an AV to implement.

(38) In practice, the CLM can handle many uncommon scenarios, but the AV will occasionally need to account for new and infrequent scenarios. For example, an AV may encounter another motorist making an abrupt and sometimes illegal U-turn. The U-turn can be at a busy intersection or could be mid-block, but the U-turn will be a sparse data point as compared to more common behaviors such as moving straight, left turns, right turns, and lane changes. Applying CLM principles, an initial deployment model may not optimally predict U-turn situations, and error situations commonly include U-turns. As the dataset grows and more error scenarios of U-turns are identified, the model can be trained to sufficiently predict a U-turn and allow the AV to accurately navigate this scenario.

(39) The CLM **200** can be applied to any number of scenarios, for example, a K-turn (or a 3-point turn), lane obstructions, construction, pedestrians, animated objects, animals, emergency vehicles, funeral processions, jaywalking, and so forth. The CLM **200** provides a mechanism for continued learning to account for diverse scenarios present in the physical world.

(40) FIG. 3 illustrates an example lifecycle **300** of an ML model in accordance with some examples. The first stage of the lifecycle **300** of an ML model is a data ingestion service **305** to generate datasets described below. ML models require a significant amount of data for the various processes described in FIG. 3 and the data persisted without undertaking any transformation to have an immutable record of the original dataset. The data itself can be generated by sensors attached to an AV, for example, but can also be provided from third-party sources such as publicly available dedicated datasets used for research purposes. The data ingestion service **305** provides a service that allows for efficient querying and end-to-end data lineage and traceability based on a dedicated pipeline for each dataset, data partitioning to take advantage of the multiple servers or cores, and spreading the data across multiple pipelines to reduce the overall time to reduce data retrieval functions.

(41) In some cases, the data may be retrieved offline that decouples the producer of the data (e.g., an AV) from the consumer of the data (e.g., an ML model training pipeline). For offline data production, when source data is available from the producer (e.g., the AV), the producer publishes a message, and the data ingestion service **305** retrieves the data. In some examples, the data ingestion service **305** may be online, and the data is streamed from the producer (e.g., the AV) in real-time for storage in the data ingestion service **305**.

(42) After data ingestion service **305**, a data preprocessing service preprocesses the data to prepare the data for use in the lifecycle **300** and includes at least data cleaning, data transformation, and data selection operations. The data preprocessing service **310** removes irrelevant data (data cleaning) and general preprocessing to transform the data into a usable form. In some examples, the data preprocessing service **310** may convert 3D LIDAR data (e.g., 3D point cloud data) into voxels. The data preprocessing service **310** includes labeling features relevant to the ML model, such as people, vegetation, vehicles, and structural objects in the case of an AV. In some examples, the data preprocessing service **310** may be a semi-supervised process performed by an ML to clean and annotate data that is complemented with manual operations such as labeling of error scenarios, identification of untrained features, etc.

(43) After the data preprocessing service **310**, data segregation service **315** separates data into at least a training dataset **320**, a validation dataset **325**, and a test dataset **330**. Each of the training dataset **320**, a validation dataset **325**, and a test dataset **330** are distinct and do not include any common data to ensure that evaluation of the ML model is isolated from the training of the ML model.

(44) The training dataset **320** is provided to a model training service **335** that uses a supervisor to perform the training or the initial fitting of parameters (e.g., weights of connections between neurons in artificial neural networks) of the ML model. The model training service **335** trains the ML model based on a gradient descent or stochastic gradient descent to fit the ML model based on an input vector (or scalar) and a corresponding output vector (or scalar). Training of the ML model

adjusts the parameters of the ML model such as weights and biases of the fully connected or convolutional neural network layers.

(45) After training, the ML model is evaluated at a model evaluation service **340** using data from the validation dataset **325** and different evaluators to tune the hyperparameters of the ML model. The predictive performance of the ML model is evaluated based on predictions on the validation dataset **325** and iteratively tunes the hyperparameters based on the different evaluators until the best fit for the ML model is identified. After the best fit is identified, the test dataset **330**, or holdout data set, is used as a final check to perform an unbiased measurement on the performance of the final ML model by the model evaluation service **340**. In some cases, the final dataset that is used for the final, unbiased measurement can be referred to as the validation dataset, and the dataset used for hyperparameter tuning can be referred to as the test dataset.

(46) After the ML model has been evaluated by the model evaluation service **340**, an ML model deployment service **345** can deploy the ML model into an application or a suitable device. The deployment can be into a further test environment such as a simulation environment or into another controlled environment to further test the ML model. In the case of an AV, the ML model may need to undergo further evaluation inside a simulated environment and, after further validation, could be deployed in the AV. In some examples, the ML model could be implemented as part of the perception stack **112** to detect objects.

(47) After deployment by the ML model deployment service **345**, a performance monitor service **350** monitors for the performance of the ML model. In some cases, the performance monitor service **350** can also record performance data such as driving data that can be ingested via the data ingestion service **305** to provide further data, additional scenarios, and further enhance the training of ML models.

(48) FIG. 4 illustrates an environment **400** having a fleet of autonomous vehicles **102-1**, **102-2**, **102-3** (collectively autonomous vehicle **102**). More specifically, each autonomous vehicle **102** can communicate with all other autonomous vehicles **102** in the fleet. Additionally, each autonomous vehicle **102** can communicate with a server or data center **150**, hosting or storing a joint kernel for a deep-learning model. As discussed above, each autonomous vehicle **102** can perceive objects in environment **400** through sensors **104-108** and perception stack **112**. For example, autonomous vehicle **102** can receive sensor data and perceive or produce objects in the sensor data. As an autonomous vehicle **102** perceives these objects and makes predictions about trajectories of the objects, autonomous vehicle **102** learns (e.g., via steps **210-260** of CLM **200**) about the accuracy of perception and predictions of the objects in environment **400**.

(49) In some embodiments, autonomous vehicle **102** has a specified learning budget that limits the total change can happen to the deep-learning model for learning. In some embodiments, the learning budget is based on a fleet size, a number of autonomous vehicles operating, a constant value, a threshold value, a type of deep-learning model, and/or a value determined by simulating a safety test. In some embodiments, the learning budget is used to calculate a learning limit, which causes autonomous vehicle **102** to reduce or stop learning after the learning limit is reached. In some embodiments, the learning limit is calculated by spreading out the learning budget over a wide time range, using a constant learning limit and setting the limit to zero when the learning budget is reached, selectively learning in certain environments, and/or selectively learning for failed perception or predictions. Additionally, the learning budget and learning limit can be used to limit how much a single autonomous vehicle changes a deep-learning model on-board the autonomous vehicle. In other words, the learning budget and learning limit can be effective to prevent excessive changes to the deep-learning model, while permitting continuous learning by autonomous vehicles **102**. Thus, the learning limit can reduce risks and errors (e.g., from small sample sizes) being introduced into the joint kernel of the deep-learning model without testing by the whole fleet. The changes to the deep-learning model due to updates can be measured by taking a norm of the difference between the original model parameters and final model parameters after

the update. The norm can be L0 (maximum), L1 (absolute) or L2 (squared). The model parameters can be weights and bias of fully connected, convolutional or any other layer of the deep neural network. The model parameter can also be any other parameter of a neural network layer such as any of the parameters of a batch-normalization layer.

(50) Autonomous vehicle **102** can calculate and/or generate a learning update to deep-learning models. After generating the learning update, autonomous vehicle **102** can apply the learning update to a kernel of the deep-learning model on-board autonomous vehicle **102**, provided that the learning limit is not reached yet. Additionally and/or alternatively, learning update may be reduced in size to make sure that the changes to the deep learning model meet the learning budget. In some embodiments, autonomous vehicle **102** can only apply the learning update to deep-learning models that are insignificant to safety or safe operation of autonomous vehicle **102**. Additionally, autonomous vehicle **102** can then perform a safety test on autonomous vehicle **102**. In other words, autonomous vehicle **102** can self-test the operations of autonomous vehicle **102** to ensure safe operation and control of autonomous vehicle **102**. In some embodiments, the learning update may be in the form of a change in the deep learning model parameters, and application of the update is adding the update on the existing deep learning model parameters. In some embodiments, the learning update may be in the form of new values for the deep learning model parameters, and application of the update is replacing the existing deep learning model parameters with the update values.

(51) In some embodiments, the safety self-test can include checks that test the accuracy of perception, predictions, tracking, planning with the deep-learning kernels for predetermined test scenarios. In some embodiments, the safety self-test can include checks that test the accuracy of perception, predictions, tracking, planning with the deep-learning kernels for the most recent sensor data received from sensors **104-108** of autonomous vehicle **102**. In some embodiments, the safety self-test can include checks verifying the difference between the outputs between the test against predetermined test sets and the most recent sensor data. After validating the safety and/or comfort of the updated operations of autonomous vehicle **102**, autonomous vehicle **102** can drive with the updated kernel for the deep-learning model.

(52) As further shown in FIG. 4, one autonomous vehicle (e.g., autonomous vehicle **102-1**) can communicate with another autonomous vehicle (e.g., autonomous vehicle **102-2**). For example, autonomous vehicle **102-1** can generate a learning update to a kernel of a deep-learning model on-board autonomous vehicle **102-1**. Autonomous vehicle **102-1** can then send the learning update to one or more of the autonomous vehicle **102-2** and data center **150**. Autonomous vehicle **102-2** can then apply the learning update to a deep-learning model on-board autonomous vehicle **102-2**. Additionally, autonomous vehicle **102-2** can also perform a safety self-test to ensure safe operations of autonomous vehicle **102-2**. After validating the safety and/or comfort of the updated operations of autonomous vehicle **102-2**, autonomous vehicle **102-2** can drive with the updated kernel for the deep-learning model.

(53) Additionally, autonomous vehicle **102-2** can generate another learning update to a kernel of a deep-learning model on-board autonomous vehicle **102-2** based on sensor data received from sensors on-board autonomous vehicle **102-2**. Both autonomous vehicle **102-1** and autonomous vehicle **102-2** can respectively send the learning update and another learning update to one or more of the autonomous vehicle **102-3** and data center **150**. Autonomous vehicle **102-3** can receive the learning update and the another learning update and generate a third learning update based on the learning update and the another learning update. In some embodiments, the third learning update can be calculated by arithmetic averaging and/or deep learning knowledge distillation. Then, autonomous vehicle **102-3** can apply the third learning update to a kernel of a deep-learning model on-board autonomous vehicle **102-3** and perform a safety self-test to ensure safe operations of autonomous vehicle **102-3**. After validating the safety and/or comfort of the updated operations of autonomous vehicle **102-3**, autonomous vehicle **102-3** can drive with the updated kernel for the

deep-learning model.

(54) In some embodiments, data center **150** can host a joint kernel that each autonomous vehicle **102** of the fleet can access and use to operate safely. The joint kernel can be updated by the learning updates generated by autonomous vehicles **102**. For example, autonomous vehicle **102-1** can send the learning update to data center **150**, which can be applied to the joint kernel. In some embodiments, the learning updates from each autonomous vehicle **102** can be averaged out or distilled (e.g., using knowledge distillation methods) and applied to the joint kernel. The joint kernel can then be sent to or retrieved by autonomous vehicles **102**. Autonomous vehicles **102** can then drive with the updated joint kernel. In some embodiments, autonomous vehicles **102** can perform a safety self-test of the autonomous vehicle **102** with the joint kernel applied. It is also contemplated that the safety self-test can be skipped when the joint kernel is verified and/or tested by data center **150**.

(55) FIG. 5 illustrates an example method **500** for sharing knowledge learned by an autonomous vehicle of a fleet. Although the example method **500** depicts a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of the method **500**. In other examples, different components of an example device or system that implements the method **500** may perform functions at substantially the same time or in a specific sequence.

(56) According to some embodiments, method **500** includes calculating **510** a learning limit based on the learning budget. For example, local computing device **110** illustrated in FIG. 1 may calculate **510** a learning limit based on the learning budget. In some embodiments, generating the update occurs after reaching the learning limit. In some embodiments, the learning budget is based on a number of autonomous vehicles in the fleet of autonomous vehicles.

(57) According to some embodiments, method **500** includes generating **520** an update for a continuous deep learning neural network on-board the autonomous vehicle based on driving scenarios encountered by the autonomous vehicle during its deployment. For example, local computing device **110** illustrated in FIG. 1 may generate **520** an update for a continuous deep learning neural network on-board the autonomous vehicle based on driving scenarios encountered by the autonomous vehicle during its deployment. In some embodiments, generating **520** an update may comprise re-training the deep neural network based additional training data which are chosen from scenarios where accuracy of the perception or predictions are low. In some embodiments, generating **520** an update may comprise applying a single or multiple back propagation steps. In some embodiments, generating **520** an update may comprise application of a numerical optimization algorithms such as gradient methods or evolutionary optimization methods.

(58) According to some embodiments, method **500** includes providing **530** the update for the continuous deep learning neural network to additional vehicles in the fleet on autonomous vehicles. For example, local computing device **110** illustrated in FIG. 1 may provide **530** the update for the continuous deep learning neural network to additional vehicles in the fleet on autonomous vehicles. In some embodiments, the update for the continuous deep learning neural network is configured to be incorporated into a joint kernel for use by the additional vehicles in the fleet on autonomous vehicles. In some embodiments, the continuous deep learning neural network is used in at least one of the perception stack, prediction stack, and planning stack.

(59) According to some embodiments, method **500** includes applying **540** the update for the continuous deep learning neural network to a kernel of the deep learning neural network on the autonomous vehicle to generate an updated kernel on the autonomous vehicle. For example, local computing device **110** illustrated in FIG. 1 may apply **540** the update for the continuous deep learning neural network to a kernel of the deep learning neural network on the autonomous vehicle to generate an updated kernel on the autonomous vehicle. In some embodiments, applying **540** the update includes at least one of averaging the update and the additional update or using knowledge

distillation on the update and the additional update

(60) According to some embodiments, method **500** includes performing **550**, by the autonomous vehicle, a safety self-test on the updated kernel. For example, local computing device **110** illustrated in FIG. **1** may perform **550**, by the autonomous vehicle, a safety self-test on the updated kernel. In some embodiments, the safety self-test includes checks that determine accuracy of the updated kernel. For example, the safety self-test can compare a performance of a deep-learning model with the updated kernel against the performance of the deep-learning model with a previous kernel.

(61) According to some embodiments, method **500** includes causing **560** the autonomous vehicle to drive with the updated kernel applied. For example, local computing device **110** illustrated in FIG. **1** may cause **560** the autonomous vehicle to drive with the updated kernel applied.

(62) FIG. **6** illustrates an example method **600** for sharing knowledge learned by an autonomous vehicle of a fleet. Although the example method **600** depicts a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of the method **600**. In other examples, different components of an example device or system that implements the method **600** may perform functions at substantially the same time or in a specific sequence.

(63) According to some embodiments, method **600** includes receiving **610** an update over a network connection for the kernel from a first other autonomous vehicle in the fleet of autonomous vehicles. For example, local computing device **110** illustrated in FIG. **1** may receive **610** an update over a network connection for the kernel from a first other autonomous vehicle in the fleet of autonomous vehicles. In some embodiments, the update received over the network connection is generated by the first other autonomous vehicle in the fleet of autonomous vehicles.

(64) According to some embodiments, method **600** includes receiving **620** an additional update over the network connection for the kernel from a second other autonomous vehicle in the fleet of autonomous vehicles. For example, local computing device **110** illustrated in FIG. **1** may receive **620** an additional update over the network connection for the kernel from a second other autonomous vehicle in the fleet of autonomous vehicles. In some embodiments, the additional update received over a network connection is generated by the second other autonomous vehicle in the fleet of autonomous vehicles.

(65) According to some embodiments, method **600** includes applying **630** the update received over the network connection from the first other autonomous vehicle and the additional update received over a network connection from the second other autonomous vehicle to the kernel of the continuous deep learning neural network to generate a second updated kernel. For example, local computing device **110** illustrated in FIG. **1** may apply **630** the update received over the network connection from the first other autonomous vehicle and the additional update received over a network connection from the second other autonomous vehicle to the kernel of the continuous deep learning neural network to generate a second updated kernel.

(66) In some embodiments, the fleet of autonomous vehicles **102** can utilize both method **500** and method **600**. For example, one autonomous vehicle can simultaneously generate a learning update (e.g., method **500**) and receive another learning update (e.g., method **600**). Additionally, one autonomous vehicle can generate a learning update (e.g., method **500**), while another autonomous vehicle in the fleet of autonomous vehicles can receive the learning update (e.g., method **600**).

(67) While the present technology has been described with respect to particular technology stacks of autonomous vehicles, it should be appreciated that the present technology can pertain to any technology stack, or even to any deployment of continuous learning models across multiple instances. Such deployments are not limited to autonomous vehicle deployments. For example, the present technology could pertain to learning of a model on a personal computing device or a cloud instance wherein the learnings can be shared and used to update other instances of the continuous

learning model on other personal computing devices or cloud instances.

(68) FIG. 7 shows an example of computing system **700**, which can be for example any computing device making up autonomous vehicle **102**, local computing device **110**, data center **150**, client computing device **170**, or any component thereof in which the components of the system are in communication with each other using connection **705**. Connection **705** can be a physical connection via a bus, or a direct connection into processor **710**, such as in a chipset architecture. Connection **705** can also be a virtual connection, networked connection, or logical connection.

(69) In some embodiments, computing system **700** is a distributed system in which the functions described in this disclosure can be distributed within a datacenter, multiple data centers, a peer network, etc. In some embodiments, one or more of the described system components represents many such components each performing some or all of the function for which the component is described. In some embodiments, the components can be physical or virtual devices.

(70) Example system **700** includes at least one processing unit (CPU or processor) **710** and connection **705** that couples various system components including system memory **715**, such as read-only memory (ROM) **720** and random access memory (RAM) **725** to processor **710**.

Computing system **700** can include a cache of high-speed memory **712** connected directly with, in close proximity to, or integrated as part of processor **710**.

(71) Processor **710** can include any general purpose processor and a hardware service or software service, such as services **732**, **734**, and **736** stored in storage device **730**, configured to control processor **710** as well as a special-purpose processor where software instructions are incorporated into the actual processor design. Processor **710** may essentially be a completely self-contained computing system, containing multiple cores or processors, a bus, memory controller, cache, etc. A multi-core processor may be symmetric or asymmetric.

(72) To enable user interaction, computing system **700** includes an input device **745**, which can represent any number of input mechanisms, such as a microphone for speech, a touch-sensitive screen for gesture or graphical input, keyboard, mouse, motion input, speech, etc. Computing system **700** can also include output device **735**, which can be one or more of a number of output mechanisms known to those of skill in the art. In some instances, multimodal systems can enable a user to provide multiple types of input/output to communicate with computing system **700**.

Computing system **700** can include communications interface **740**, which can generally govern and manage the user input and system output. There is no restriction on operating on any particular hardware arrangement, and therefore the basic features here may easily be substituted for improved hardware or firmware arrangements as they are developed.

(73) Storage device **730** can be a non-volatile memory device and can be a hard disk or other types of computer readable media which can store data that are accessible by a computer, such as magnetic cassettes, flash memory cards, solid state memory devices, digital versatile disks, cartridges, random access memories (RAMs), read-only memory (ROM), and/or some combination of these devices.

(74) The storage device **730** can include software services, servers, services, etc., that when the code that defines such software is executed by the processor **710**, it causes the system to perform a function. In some embodiments, a hardware service that performs a particular function can include the software component stored in a computer-readable medium in connection with the necessary hardware components, such as processor **710**, connection **705**, output device **735**, etc., to carry out the function.

(75) For clarity of explanation, in some instances, the present technology may be presented as including individual functional blocks including functional blocks comprising devices, device components, steps or routines in a method embodied in software, or combinations of hardware and software.

(76) Any of the steps, operations, functions, or processes described herein may be performed or implemented by a combination of hardware and software services or services, alone or in

combination with other devices. In some embodiments, a service can be software that resides in memory of a client device and/or one or more servers of a content management system and perform one or more functions when a processor executes the software associated with the service. In some embodiments, a service is a program or a collection of programs that carry out a specific function. In some embodiments, a service can be considered a server. The memory can be a non-transitory computer-readable medium.

(77) In some embodiments, the computer-readable storage devices, mediums, and memories can include a cable or wireless signal containing a bit stream and the like. However, when mentioned, non-transitory computer-readable storage media expressly exclude media such as energy, carrier signals, electromagnetic waves, and signals per se.

(78) Methods according to the above-described examples can be implemented using computer-executable instructions that are stored or otherwise available from computer-readable media. Such instructions can comprise, for example, instructions and data which cause or otherwise configure a general purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions. Portions of computer resources used can be accessible over a network. The executable computer instructions may be, for example, binaries, intermediate format instructions such as assembly language, firmware, or source code. Examples of computer-readable media that may be used to store instructions, information used, and/or information created during methods according to described examples include magnetic or optical disks, solid-state memory devices, flash memory, USB devices provided with non-volatile memory, networked storage devices, and so on.

(79) Devices implementing methods according to these disclosures can comprise hardware, firmware and/or software, and can take any of a variety of form factors. Typical examples of such form factors include servers, laptops, smartphones, small form factor personal computers, personal digital assistants, and so on. The functionality described herein also can be embodied in peripherals or add-in cards. Such functionality can also be implemented on a circuit board among different chips or different processes executing in a single device, by way of further example.

(80) The instructions, media for conveying such instructions, computing resources for executing them, and other structures for supporting such computing resources are means for providing the functions described in these disclosures.

Claims

1. A computer-implemented method for enabling sharing of knowledge learned by an autonomous vehicle to a fleet of autonomous vehicles, the method comprising: generating an update for a continuous deep learning neural network on-board the autonomous vehicle based on driving scenarios encountered by the autonomous vehicle during its deployment; providing the update for the continuous deep learning neural network to additional vehicles in the fleet on autonomous vehicles, wherein the update for the continuous deep learning neural network is configured to be incorporated into a joint kernel for use by the additional vehicles in the fleet on autonomous vehicles; and causing the autonomous vehicle to drive with the updated kernel applied; and determining a learning budget wherein the learning budget is a maximum total change to the deep-learning neural network and calculating a learning limit by evenly distributing the learning budget across a me range, using the learning limit and setting the constant learning limit to zero when the learning budget is reached, wherein generating update occurs after reaching the learning limit.
2. The computer-implemented method of claim 1, further comprising: applying the update for the continuous deep learning neural network to a kernel of the deep learning neural network on the autonomous vehicle to generate an updated kernel on the autonomous vehicle; and performing, by the autonomous vehicle, a safety self-test on the updated kernel.
3. The computer-implemented method of claim 2, wherein the safety self-test includes checks that

determine accuracy of the updated kernel.

4. The computer-implemented method of claim 1, wherein the continuous deep learning neural network is used in at least one of the perception stack, prediction stack, and planning stack.

5. The computer-implemented method of claim 1, further comprising: receiving an update over a network connection for the kernel of the continuous deep learning neural network from another autonomous vehicle in the fleet of autonomous vehicles; applying the update received over the network connection to the kernel of the continuous deep learning neural network to generate a second updated kernel; and performing, by the autonomous vehicle, a safety self-test on the second updated kernel.

6. The computer-implemented method of claim 1, wherein the learning limit is based on at least one of a fleet size, a number of autonomous vehicles operating, a constant value, a threshold value, a type of deep-learning model, and a value determined by simulating a safety test and wherein the learning limit limits how much a single autonomous vehicle changes the deep learning neural network on-board the autonomous vehicle.

7. The computer-implemented method of claim 6, wherein the learning budget is based on a number of autonomous vehicles in the fleet of autonomous vehicles.

8. The computer-implemented method of claim 1, further comprising: receiving an update over a network connection for the kernel from a first other autonomous vehicle in the fleet of autonomous vehicles, wherein the update received over the network connection is generated by the first other autonomous vehicle in the fleet of autonomous vehicles; receiving an additional update over the network connection for the kernel from a second other autonomous vehicle in the fleet of autonomous vehicles, wherein the additional update received over a network connection is generated by the second other autonomous vehicle in the fleet of autonomous vehicles; and applying the update received over the network connection from the first other autonomous vehicle and the additional update received over a network connection from the second other autonomous vehicle to the kernel of the continuous deep learning neural network to generate a second updated kernel.

9. The computer-implemented method of claim 8, wherein applying the update comprises one of averaging the update and the additional update or using knowledge distillation on the update and the additional update.

10. A system comprising: a storage configured to store instructions; a processor configured to execute the instructions and cause the processor to: generate an update for a continuous deep learning neural network on-board the autonomous vehicle based on driving scenarios encountered by the autonomous vehicle during its deployment, provide the update for the continuous deep learning neural network to additional vehicles in the fleet on autonomous vehicles, wherein the update for the continuous deep learning neural network is configured to be incorporated into a joint kernel for use by the additional vehicles in the fleet on autonomous vehicles; and cause the autonomous vehicle to drive with the updated kernel applied; and determining a learning budget, wherein the learning budget is a maximum total change to the deep-learning neural network and calculating a learning limit by evenly distributing the learning budget across a time range, using the learning limit and setting the constant learning limit to zero when the learning budget is reached, wherein generating the update occurs after reaching the learning limit.

11. The system of claim 10, wherein the processor is configured to execute the instructions and cause the processor to: apply the update for the continuous deep learning neural network to a kernel of the deep learning neural network on the autonomous vehicle to generate an updated kernel on the autonomous vehicle; and perform, by the autonomous vehicle, a safety self-test on the updated kernel.

12. The system of claim 10, wherein the continuous deep learning neural network is used in at least one of the perception stack, prediction stack, and planning stack.

13. The system of claim 10, wherein the processor is configured to execute the instructions and

cause the processor to: receive an update over a network connection for the kernel of the continuous deep learning neural network from another autonomous vehicle in the fleet of autonomous vehicles; apply the update received over the network connection to the kernel of the continuous deep learning neural network to generate a second updated kernel; and perform, by the autonomous vehicle, a safety self-test on the second updated kernel.

14. The system of claim 10, wherein the learning limit is based on at least one of a fleet size, a number of autonomous vehicles operating, a constant value, a threshold value, a type deep-learning model, and a value determined by simulating a safety test, and wherein the learning limit limits how much a single autonomous vehicle changes the deep learning neural network on-board the autonomous vehicle.

15. The system of claim 10, wherein the processor is configured to execute the instructions and cause the processor to: receive an update over a network connection for the kernel from a first other autonomous vehicle in the fleet of autonomous vehicles, wherein the update received over the network connection is generated by the first other autonomous vehicle in the fleet of autonomous vehicles; receive an additional update over the network connection for the kernel from a second other autonomous vehicle in the fleet of autonomous vehicles, wherein the additional update received over a network connection is generated by the second other autonomous vehicle in the fleet of autonomous vehicles; and apply the update received over the network connection from the first other autonomous vehicle and the additional update received over a network connection from the second other autonomous vehicle to the kernel of the continuous deep learning neural network to generate a second updated kernel.

16. A non-transitory computer readable medium comprising instructions, the instructions, when executed by a computing system, cause the computing system to: generate an update for a continuous deep learning neural network on-board the autonomous vehicle based on driving scenarios encountered by the autonomous vehicle during its deployment; provide the update for the continuous deep learning neural network to additional vehicles in the fleet on autonomous vehicles, wherein the update for the continuous deep learning neural network is configured to be incorporated into a joint kernel for use by the additional vehicles in the fleet on autonomous vehicles; and cause the autonomous vehicle to drive with the updated kernel applied, and determining learning budget, wherein the learning budget is a maximum total change to the deep-learning neural network and calculating learning limit by evenly distributing the learning budget across a time range, using the learning limit and setting the constant learning limit to zero when the learning budget is reached, wherein generating the update occurs after reaching the learning limit.

17. The computer readable medium of claim 16, wherein the computer readable medium further comprises instructions that, when executed by the computing system, cause the computing system to: apply the update for the continuous deep learning neural network to a kernel of the deep learning neural network on the autonomous vehicle to generate an updated kernel on the autonomous vehicle; and perform, by the autonomous vehicle, a safety self-test on the updated kernel.

18. The computer readable medium of claim 16, the continuous deep learning neural network is used in at least one of the perception stack, prediction stack, and planning stack.

19. The computer readable medium of claim 16, wherein the computer readable medium further comprises instructions that, when executed by the computing system, cause the computing system to: receive an update over a network connection for the kernel of the continuous deep learning neural network from another autonomous vehicle in the fleet of autonomous vehicles; apply the update received over the network connection to the kernel of the continuous deep learning neural network to generate a second updated kernel; and perform, by the autonomous vehicle, a safety self-test on the second updated kernel.

20. The computer readable medium of claim 16, wherein the learning limit is based on at least one of a fleet size, a number of autonomous vehicles operating, a constant value, a threshold value, a

type of deep-learning model, and a value determined by simulating a safety test and wherein the learning limit limits how much a single autonomous vehicle changes the deep learning neural network on-board the autonomous vehicle.
